Step 6 : Decision

As the calculated value $\chi_0^2 = 22.152$ is greater than the critical value $\chi^2_{2,0.05} = 5.991$, the null hypothesis H_0 is rejected. Hence, the performance and age of students are not independent.

NOTE

If the contigency table is 2 x 2 then the value of χ^2 can be calculated as given below:

	A	not A	Total
B	a	b	$a+b$
not B	c	d	$c+d$
Total	$a+c$	$b+d$	$N=a+b+c+d$

$$\chi_0^2 = \frac{N(ad-bc)^2}{(a+b)(c+d)(a+c)(b+d)} \sim \chi_\alpha^2(1d.f)$$

The following example will illustrate the procedure

Example 2.10

A survey was conducted with 500 female students of which 60% were intelligent, 40% had uneducated fathers, while 30 % of the not intelligent female students had educated fathers. Test the hypothesis that the education of fathers and intelligence of female students are independent.

Solution:

Step 1 : Null hypothesis H_0: The attributes are independent *i.e.* No association between education fathers and intelligence of female students

Alternative hypothesis H_1: The attributes are not independent *i.e* there is association between education of fathers and intelligence of female students

Step 2 : Data

The observed frequencies (O) has been computed from the given information as under.

	Intelligent females	Not intelligent females	Row total
Educated fathers	$300-120=180$	$\frac{30}{100}\times 200 = 60$	240
Uneducated fathers	$\frac{40}{100}\times 300 = 120$	$200-60=140$	260
Total	$\frac{60}{100}\times 500 = 300$	$500-300=200$	N= 500

Step 3 : Level of significance

$\alpha = 5\%$

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